Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
NOVEMBER/DECEMBER 2024
SUBJECT: CIPHER SYSTEM (ECE 4069), PE-V, SET-2

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	M*	C*
1A.	Solve for x, given 314x=1 mod 841	3	1
1B.	Using primitive polynomial $p(x)=1+x+x^4$, Construct the GF (2^4). Also Determine i. Sum of α^6, α^{10} ii. Product of $(1+\alpha^2+\alpha^{13})$, $(1+\alpha+\alpha^3)$ iii. Inverse of $(\alpha^2+\alpha^3$	3	1
1C.	Solve using repeated squaring technique 125^13 mod 841.	4	1
2A.	Determine the cipher text for the plain text **EXAMS.** using affine cipher with encryption key (23,10), Use the 29 letter alphabet {A,…Z,'?',,'dot', 'blankspace'} with numerical values (0-28) respectively.	5	2
2B.	Determine the plain text **TEJL** using the Hill cipher with key [9 7; 6 25]. Consider 26-letter alphabet, A-Z with numerical values 0-25 respectively.	3	2
2C.	Determine the Playfair matrix to encrypt plain text "**DO NOT COPY**", with Key: **EXAM**.	2	3
3A.	Using S-DES cryptosystem, determine the **plain text** from the intercepted message "01110100" with encryption key: (1100110011)	5	3
3B.	Calculate the Inverse Sub Byte transformation of 0xCD. Show the computation steps clearly. Verify the result using the table.	3	3
3C.	Explain the type of DES Key Generation algorithm.	2	3
4A.	Alice and Bob use Merkle Hellman Knapsack cryptosystem. , Determine the public keys of Alice, whose uses private key (40, 213)	5	4

	and sequence {2,5,15,40,65}. Determine the cipher text sent by Bob to Alice for the plain text HIDE. Use the 26 letter alphabet 26-letter alphabet, A-Z with numerical values 0-25 respectively to represent the plain text.		
4B.	Alice applies RSA algorithm to send plain text to Bob. Bob s public key is (17, 209). Determine the cipher text for the plain text "SECURE" Use the 26 letter alphabet (A,…,Z) with numerical values (0,1….25) respectively.	3	4
4C.	In the elliptic curve group defined by $y^2 = x^3 - 17x + 16$ over real numbers, Determine i. the sum of of P, Q if P = (0,-4) and Q = (1, 0)	2	4
5A.	An elliptic curve is defined by $E_{97}(3,21)$. Determine the 4P if point P is (2,36).	5	5
5B.	Explain the working of SHA512. Show the computation of values of constants in message digest initialization.	3	5
5C.	Explain the MAC requirements and properties.	2	5

M*--Marks, C*--CLO, A*--AHEP LO, B* Blooms Taxonomy Level